

Mapping data practices in gaming across publishing platforms, community sites, and online multiplayer games

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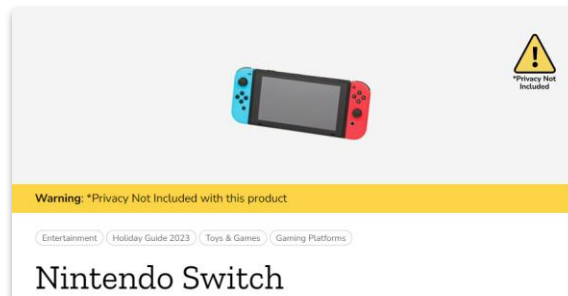
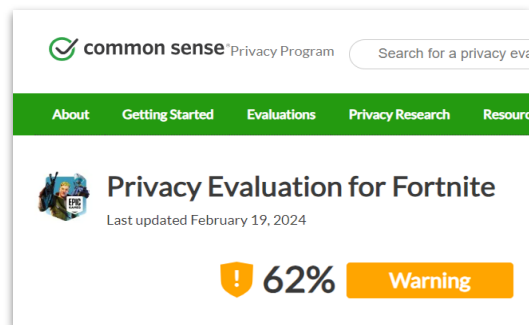
Approaching gaming as a form of social media

- The number of gamers worldwide is estimated to be over 3 billion (Newzoo 2023).
- Some surveys show 33% of adults aged 18 to 44 playing video games as leisure activity in a typical week (YouGov 2023).
- Gaming is part of the ambience of everyday life and an increasingly common form of social interaction (Richardson and Hjorth 2020).

“Before Instagram, Facebook, Twitter, and all the other platforms that we now umbrella under the term ‘social media’, sites for real-time social interaction via the internet already existed - computer games.” (Taylor 2023: 1)

What are common data practices in gaming?

- Digital platforms collect large amounts of data to enhance the experiences they offer.
- User trust is not always given. Distrust harms experiences and online self-expressions (e.g. Kappeler et al. 2023).
- Scholarship on privacy is growing. However, focussed research on trust and privacy in gaming is still limited (see Bourdouden et al. 2023).
- Moreover, gaming is rarely discussed alongside the likes of Facebook in larger social media privacy debates.



Epic Games to Pay \$520 Million Over Children's Privacy and Trickery Charges

The creator of Fortnite and other popular games violated children's privacy and duped millions of users into unwanted purchases, federal regulators said.

Games > FPS > Valorant

The controversy over Riot's Vanguard anti-cheat software, explained

Features By Tyler Wilde published 8 May 2020

Does Riot's always-on anti-cheat go too far?

Our objective: gaining a foundational understanding for trust and data literacy in the gaming context



**Landscape
survey to
understand
common data
practices**



Platform
walkthroughs as
well as technical
evaluation of key
data practices



Qualitative
investigation on
experience of
data practices
amongst gamers



Quantitative study
to evaluate levels
of awareness,
trust, and literacy
amongst gamers

Three types of platforms surveyed in first project stage

Publishing Platforms

... defined as the platforms on and through which people access games. Access here meaning purchasing games, managing libraries of game contents, downloading and installing game content to local devices, launching games through the platform clients.

Cases: Apple App Store, Epic Store, Google Play Store, Nintendo, PlayStation, Riot Games, Roblox, Steam, WeGame, Xbox.

Online Games

... defined as video games that are played online with and versus other players. Due to that, these games often involve the use of data intensive technologies, for example for social interaction, community moderation, or creating fair play conditions.

Cases: Apex Legends, Brawl Stars, Call of Duty, Clash of Clans, Counter-Strike, DotA2, FIFA (EA FC), Fortnite, League of Legends, Minecraft, Mobile Legends, PUBG, Roblox, Rocket League, Valorant.

Community Sites

... defined as the platforms on which people connect as gamers with other gamers. This can include interactions with friends, but also public online communities, as well as through content and creators on other social media platforms.

Cases: Discord, Douyu, FACEIT, Facebook, Huya, KOOK, PerfectWorld Arena, Reddit, Twitch, YY, YouTube Gaming.



Four main thematic groups at a glance

Consent, Control, Security

Practices relating to the management of data and collection through the platform, including the positioning of users as data subjects.

Platform Experience

Practices concerned with using data to enhance and modulate the users experience on the platform, for example through personalisation.

Marketing and Business Operations

Practices in which data is used for a variety of business related purposes such as marketing, advertising, or research analytics.

Third-party Operations

Practices in which data interactions involve third-parties, mostly advertising but also other platforms, creators, and users.

Findings: Consent, Control, Security

Automatic collection

- Device and system
- Platform usage
- Gameplay data
- Purchase info
- Ingame actions
- Cookies info
- Other tracking
- Voice and text chat
- Social interaction
- User content
- Geolocation

Opt-in collection

- Personal
- Account
- Payment
- Health
- Physical
- Survey data
- Underage users

User controls

- Adjust settings
- Update and correct
- Delete data / account
- Data access requests
- Contact platform
- Local subject rights
- Right to portability
- External privacy tools
- Legal representation

Security

- Framing security expectations
- Technical data security management
- Organisational data security management



Findings: Marketing and Business Operations

Marketing and Advertising

- Opt-in communications
- Localized promotions
- Cookies personalisation
- Predictive analytics
- Marketing measurement
- Internal data aggregation
- Third-party sources

Customer relations and Support

- Processing personal info
- Accessing technical info
- Verification of identities
- Storing interactions

Research and Analytics

- Analysing features
- User needs and trends
- Training data models
- Data driven features
- Analytics cookies
- Product reviews
- Feedback polls
- Rating pop ups
- Research surveys

Management and Legal

- Analysing and reporting on business performance
- Data transfer in case of sale of platform
- Data transfer inside a corporate group
- Responding to law enforcement requests
- Compliance and safety duties (e.g. mitigate harm)
- Retaining data for potential legal disputes

Findings: Platform Experience

Personalisation

- Content personalisation (e.g., based on usage, interests, localisation)
- Personalisation of interactions (e.g. game matchmaking systems)
- Friends and social recommendations (e.g. communities to join)
- Basic customisation, localization, and storage of preferences (e.g. via cookies technologies)

Moderation

- Anti-cheat systems for fair play conditions
- Various fraud detection (e.g. payment related)
- Anti-toxicity systems for ingame moderation
- Content detection, filtering, and moderation
- User reports and complaint processing
- Tools for community moderators

Technical maintenance

- Improve and deliver the “expected” experience
- Maintain the service on the user device
- Device related information to enable automatic updates
- Errors reports for troubleshooting and system safety

Platform access

- Login via personal and account information
- Checking technical device and system identifiers
- First party cookies and similar technologies
- Verifying products, licenses, subscriptions, etc

Findings: Third-Party Operations

Sharing data

- Sharing identifiers with third parties (e.g., for advertising purposes)
- Sharing data with developers of games hosted on the platform
- Sharing data for ads delivery and performance measurement
- Sharing data and content with services providers (e.g. data centres)
- Sharing personal information for service delivery (e.g. payment)

Receiving data

- Account linkage info (phone, social, gaming)
- Data from sites that use a platform's cookies
- Receiving demographics from data brokers
- Information from banks and financial institutions
- Information about device (e.g. fraud scores)
- General cookie tracking data from partners
- Data from partners and affiliates

Collecting data

- Games collecting data through platform
- Developers of features collecting users info
- Cookies and embedded content that collects data
- Users recording gameplay and communication

Service providers

- Platform experience and content (e.g. moderation services, creators, etc.)
- Commercial processing partners (e.g. payment services, tournament organisers, etc.)
- Media and advertising services (e.g. ads measurement, etc.)
- Technology related services (e.g. data centre and hosting services, etc.)
- Business services (e.g. customer support, user acquisition tools, etc.)



Some conclusions after surveying the landscape

- There is homogeneity in the data practices of gaming platforms.
- There are strong commonalities with other forms of social media.
- Terms of data collected, especially ingame related, are often opaque.
- Gaming is a data-intensive and data-reliant leisure activity.
- “Gamer” can be grasped as a key contemporary data subject position.

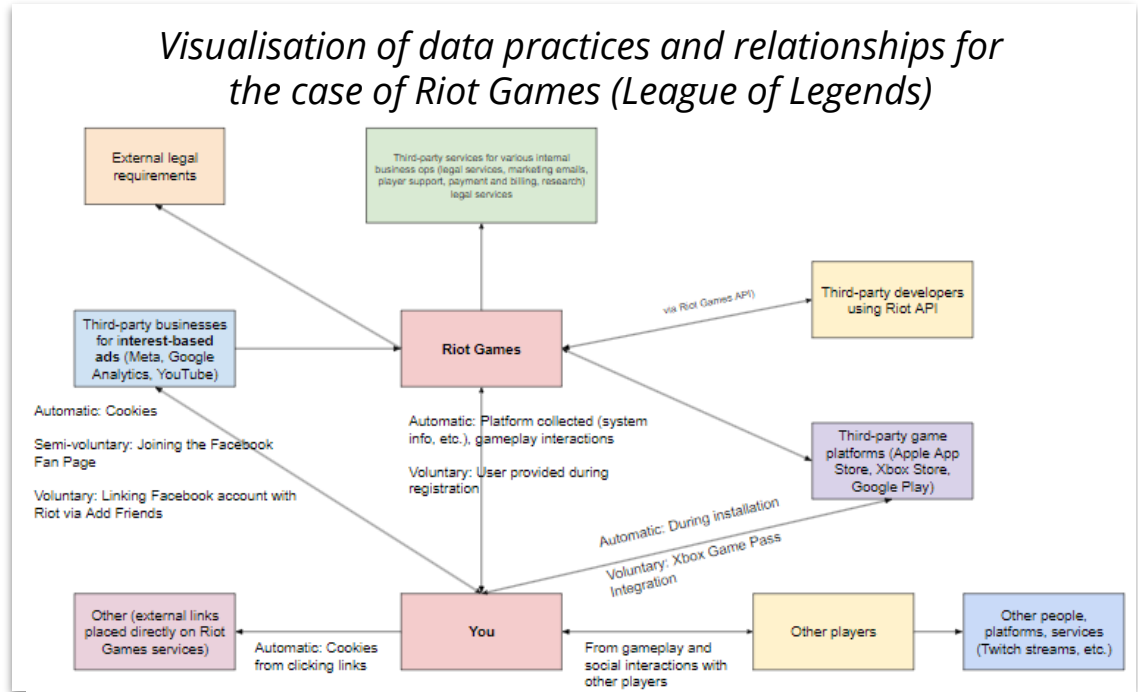
Areas of further exploration and research

Based on our findings we...

- suspect possible issues in meaningful consent being established, especially across platforms.
- identify anti-cheat, anti-toxicity, and game matchmaking as data practices relatively unique to gaming context.
- see a strong integration of gaming into advertising networks and data aggregation processes (both as data source and identifier).

Qualitatively explore gamers' experiences

- Understanding trust as a relational construct (Sherman 2020), we are going to further investigate how gamers understand their position in the gaming landscape.
- Particularly, the data-reliant nature of gaming as a practice poses the question of how gamers negotiate their position of potential vulnerability in the trust relationship (Baier 1986).



Thanks!

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